

Gebze Technical University
Dep. of Mechanical Engineering

Statics (ME 231)
Lecture 2: Force Vectors
(Geometric Analysis)
By Saeed Lotfan

Force Vectors (Geometric Analysis)

Scalars and Vectors:

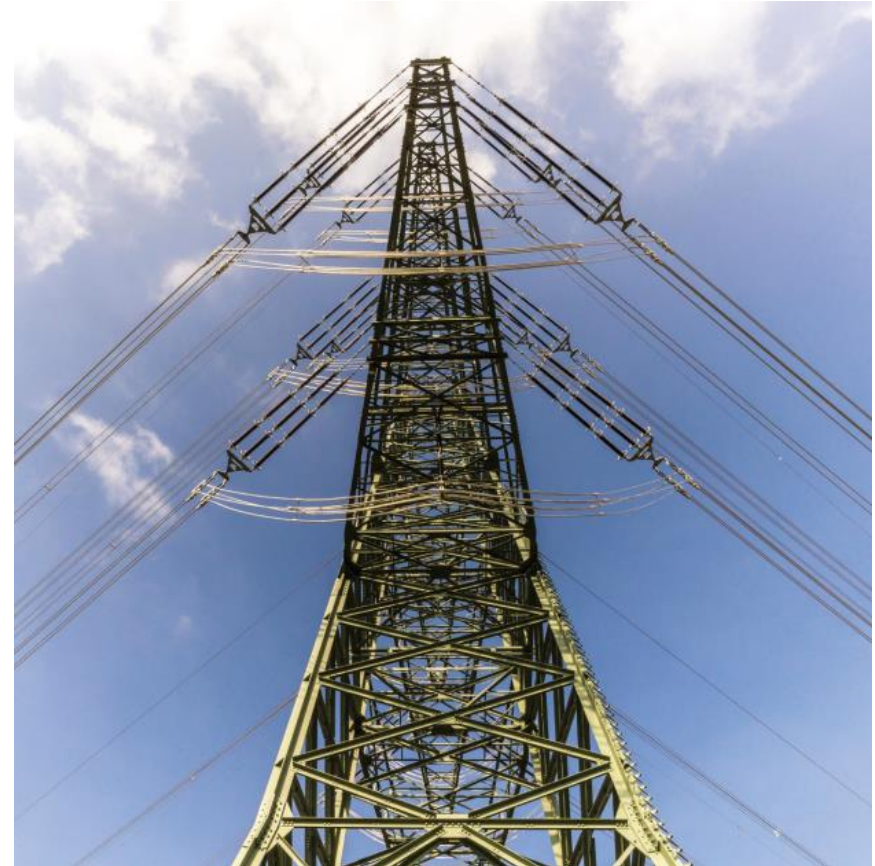
- Scalar

magnitude a

- Vector

magnitude + direction $\vec{a}$

+ obeying rules of vectors.

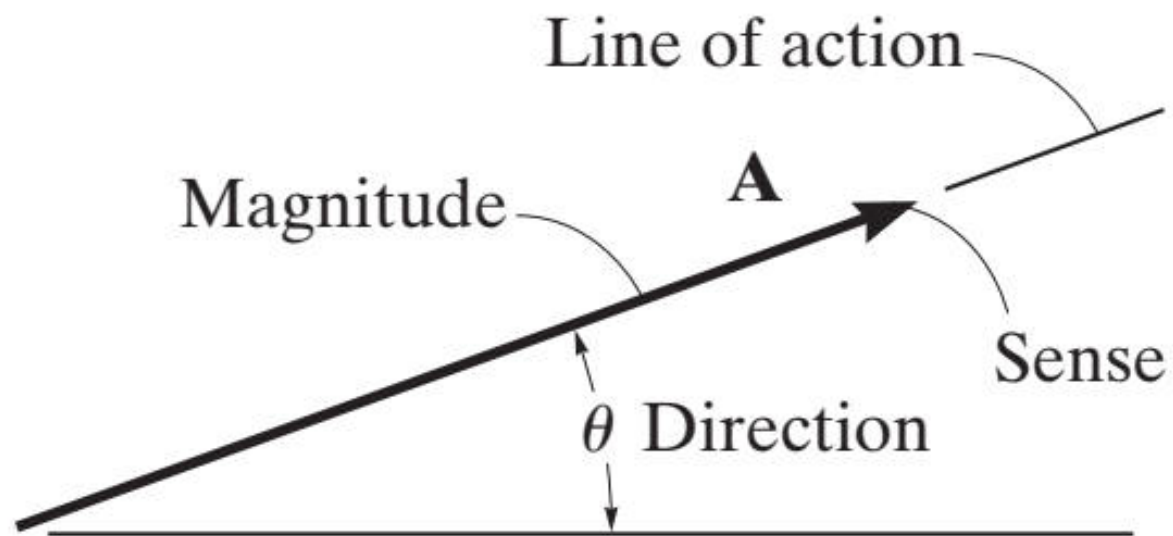


How to know that a parameter is a vector in the context of a book?

bold small letters $\{a\}$, $\vec{a}$

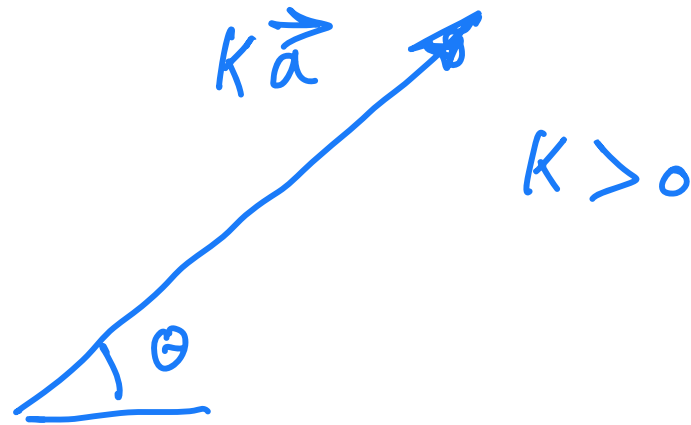
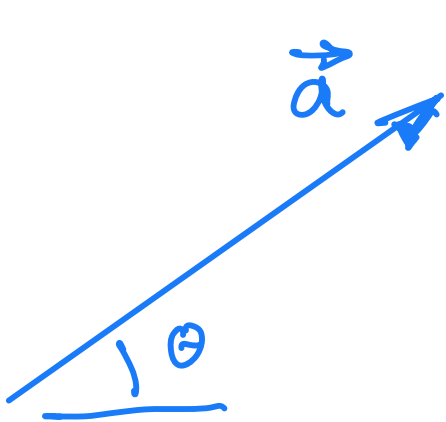
Force Vectors (Geometric Analysis)

Scalars and Vectors:



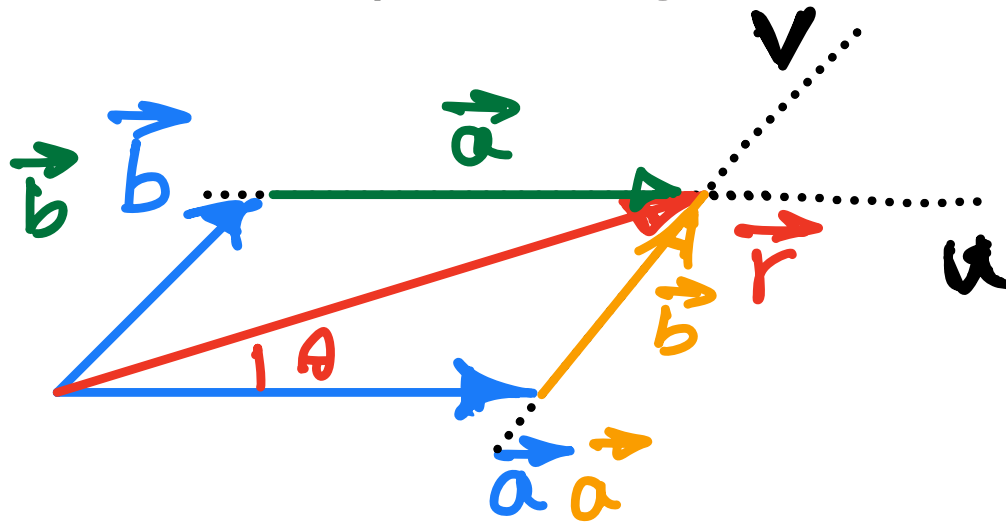
Vector Operations

Multiplication and Division of a Vector by a Scalar:



Vector Operations

Vector Addition (*parallelogram law of addition*):



$$\vec{r} = \vec{a} + \vec{b}$$

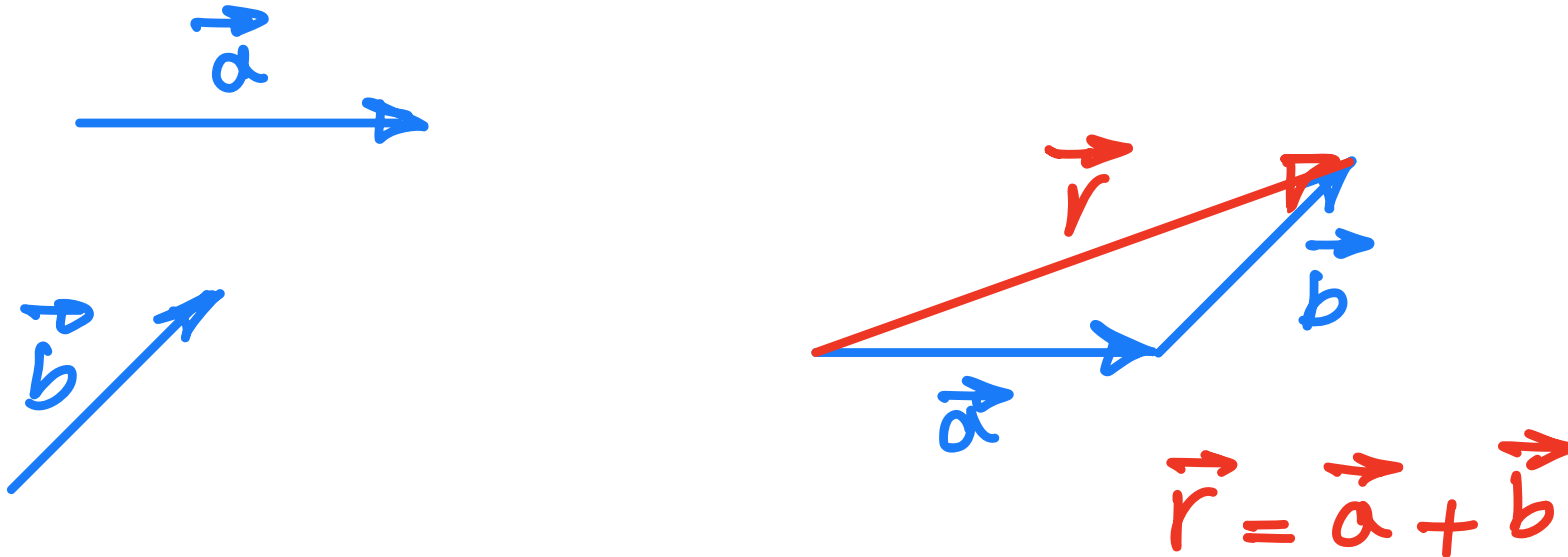
$$\vec{r} = \vec{b} + \vec{a}$$

$$\vec{b} + \vec{a} = \vec{a} + \vec{b} = \vec{r}$$

Make sure you understand the component and resultant vectors concepts.

Vector Operations

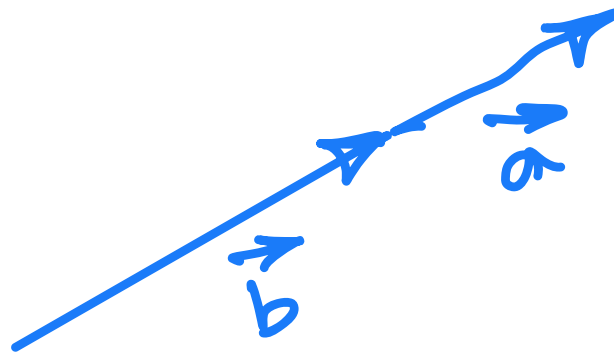
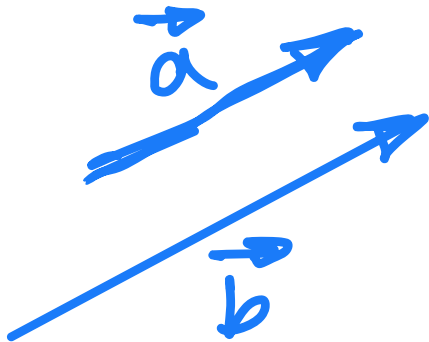
Vector Addition (*triangle rule*):



Make sure you understand the commutative nature of vector additions.

Vector Operations

Vector Addition (*collinear case*):



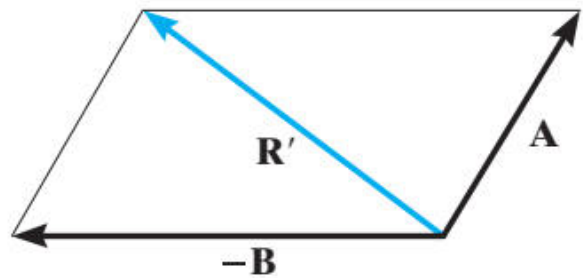
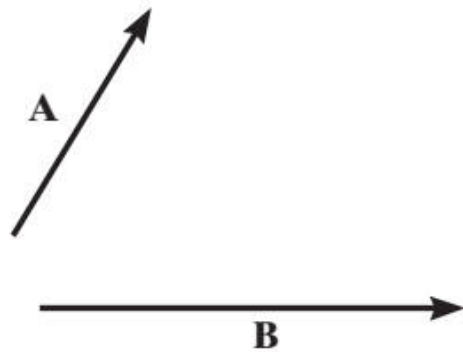
$$\vec{r} = \vec{b} + \vec{a}$$

$$r = b + a$$

Make sure you see the algebraic addition for the magnitudes.

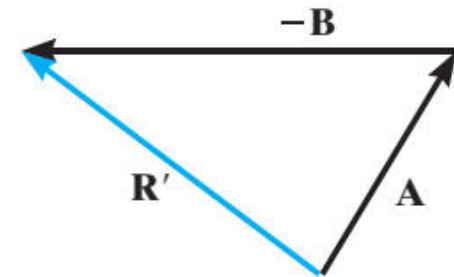
Vector Operations

Vector Subtraction (*parallelogram law of addition and triangle rule*)



Parallelogram law
Vector subtraction

or



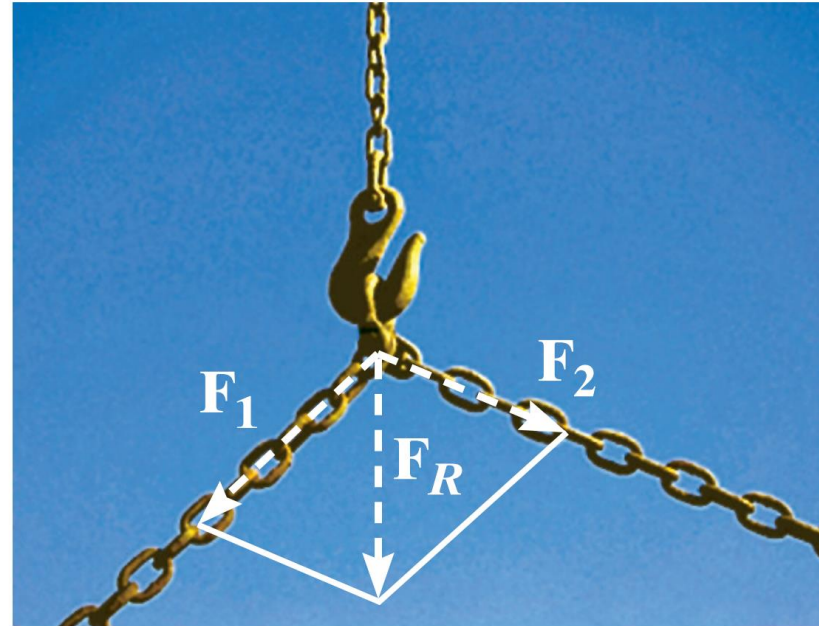
Triangle construction

What about collinear case?

Vector Addition of Forces

Experimental evidence has shown that a force is a vector quantity since it has a specified magnitude, direction, and sense and it adds according to the parallelogram law.

Finding a Resultant Force:

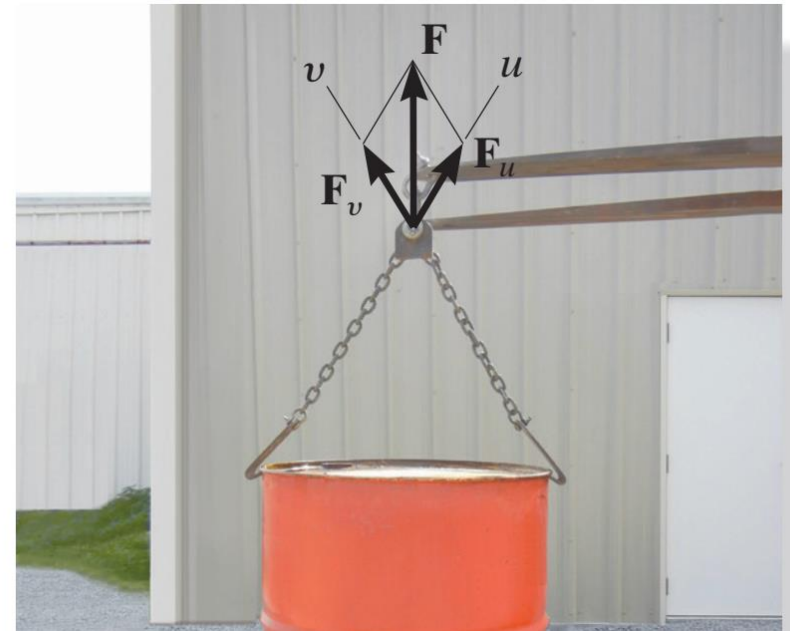
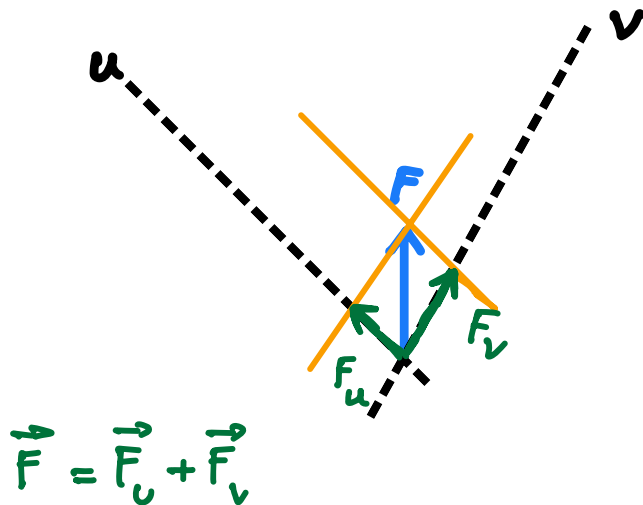


The parallelogram law must be used to determine the resultant of the two forces acting on the hook.

Vector Addition of Forces:

Finding the Components of a Force

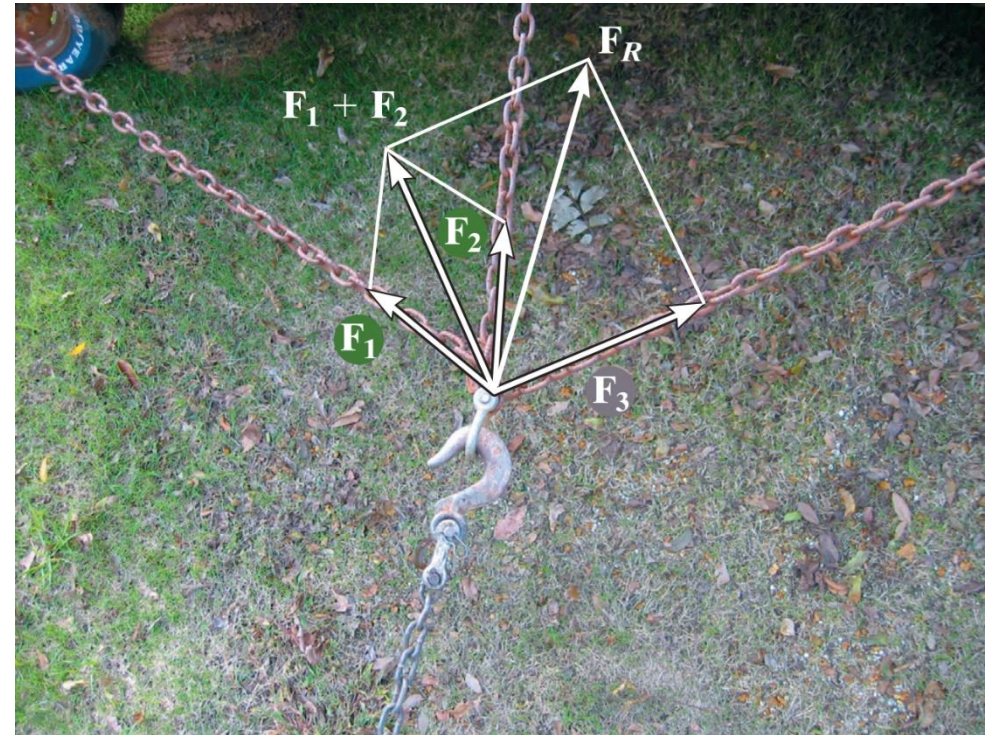
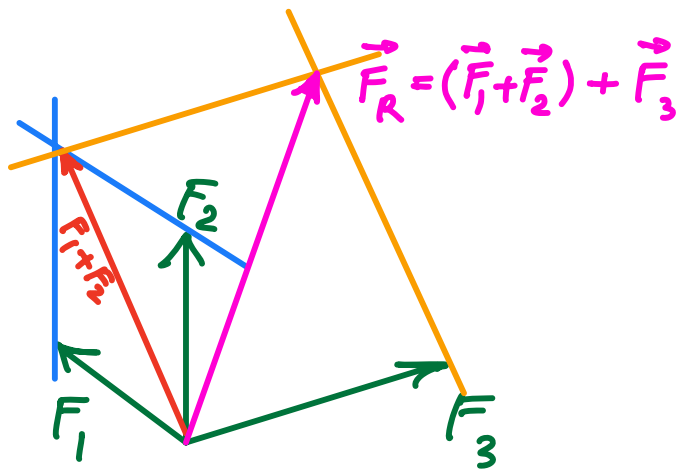
Sometimes it is necessary to resolve a force into two components in order to study its pulling or pushing effect in two specific directions.



Using the parallelogram law the supporting force $\mathbf{F}$ can be resolved into components acting along the u and v axes.

Vector Addition of Forces

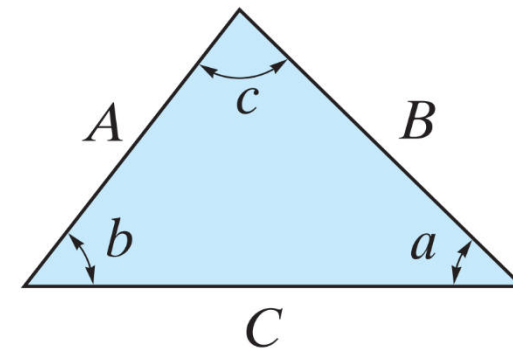
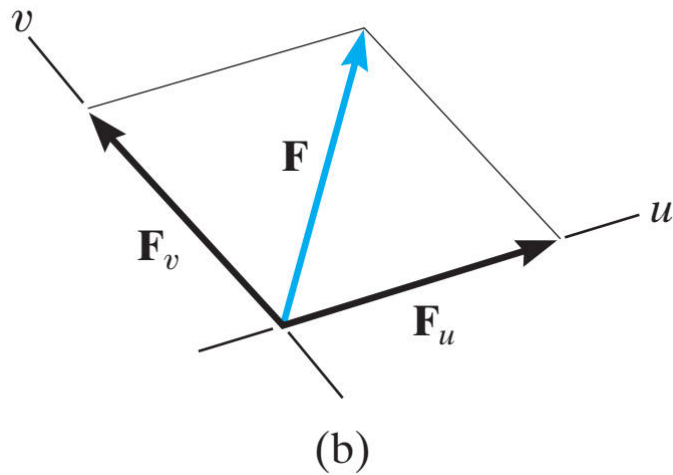
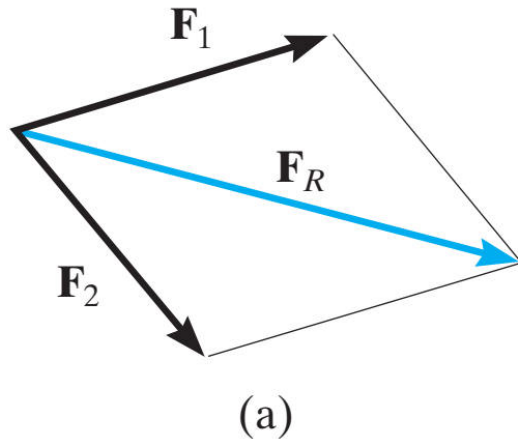
Addition of Several Forces



Concluding Notes

- A scalar is a positive or negative number.
- A vector is a quantity that has a magnitude, direction, and sense.
- Multiplication or division of a vector by a scalar will change the magnitude of the vector. The sense of the vector will change if the scalar is negative.
- Vectors are added or subtracted using the parallelogram law or the triangle rule.
- As a special case, if the vectors are collinear, the resultant is formed by an algebraic or scalar addition.

Notes for Geometric Analysis



Cosine law:

$$C = \sqrt{A^2 + B^2 - 2AB \cos c}$$

Sine law:

$$\frac{A}{\sin a} = \frac{B}{\sin b} = \frac{C}{\sin c}$$

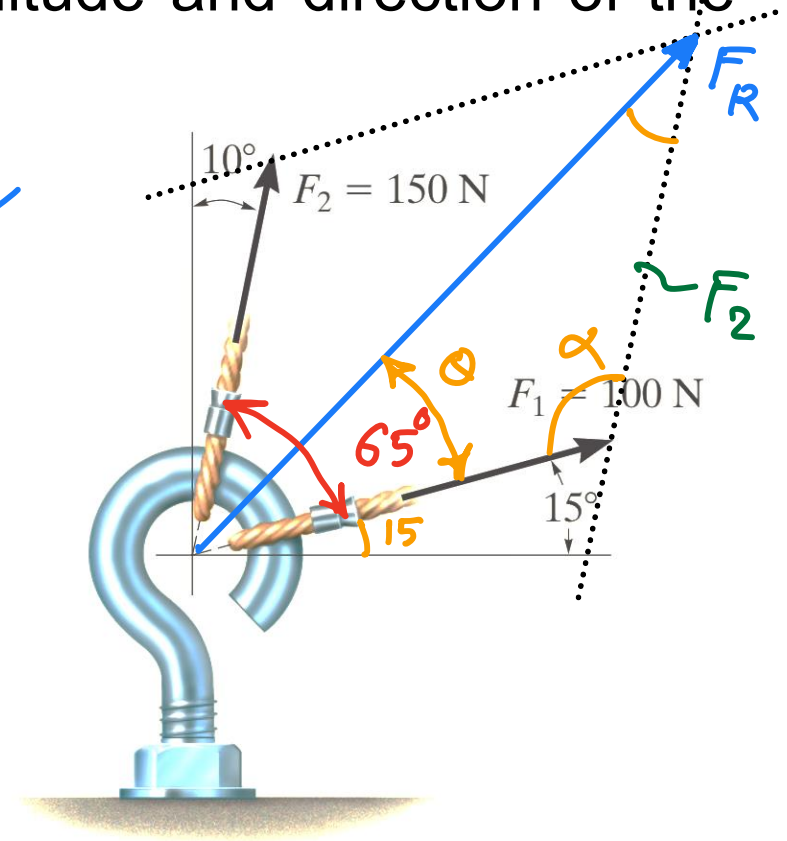
(c)

Example: The screw eye in the figure below is subjected to two forces, $\mathbf{F}_1$ and $\mathbf{F}_2$. Determine the magnitude and direction of the resultant force.

$$F_R^2 = F_1^2 + F_2^2 + 2F_1F_2\cos 65 \rightarrow F_R \checkmark$$

$$F_R^2 = F_1^2 + F_2^2 - 2F_1F_2\cos \alpha \rightarrow \alpha$$

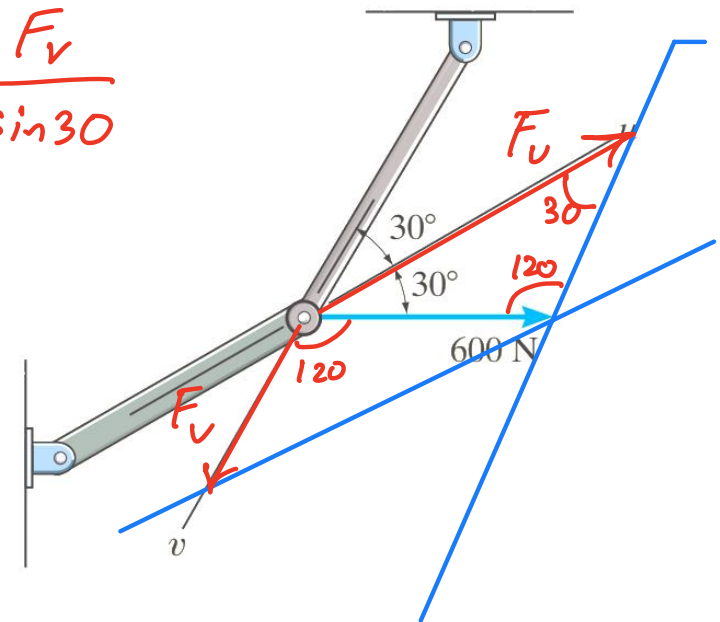
$$\frac{F_R}{\sin \alpha} = \frac{F_2}{\sin \theta}$$



Practice at Home

1. Resolve the horizontal 600-N force in the figure below into components acting along the u and v axes and determine the magnitudes of these components.

$$\frac{F_u}{\sin 120} = \frac{F}{\sin 30} = \frac{F_v}{\sin 30}$$

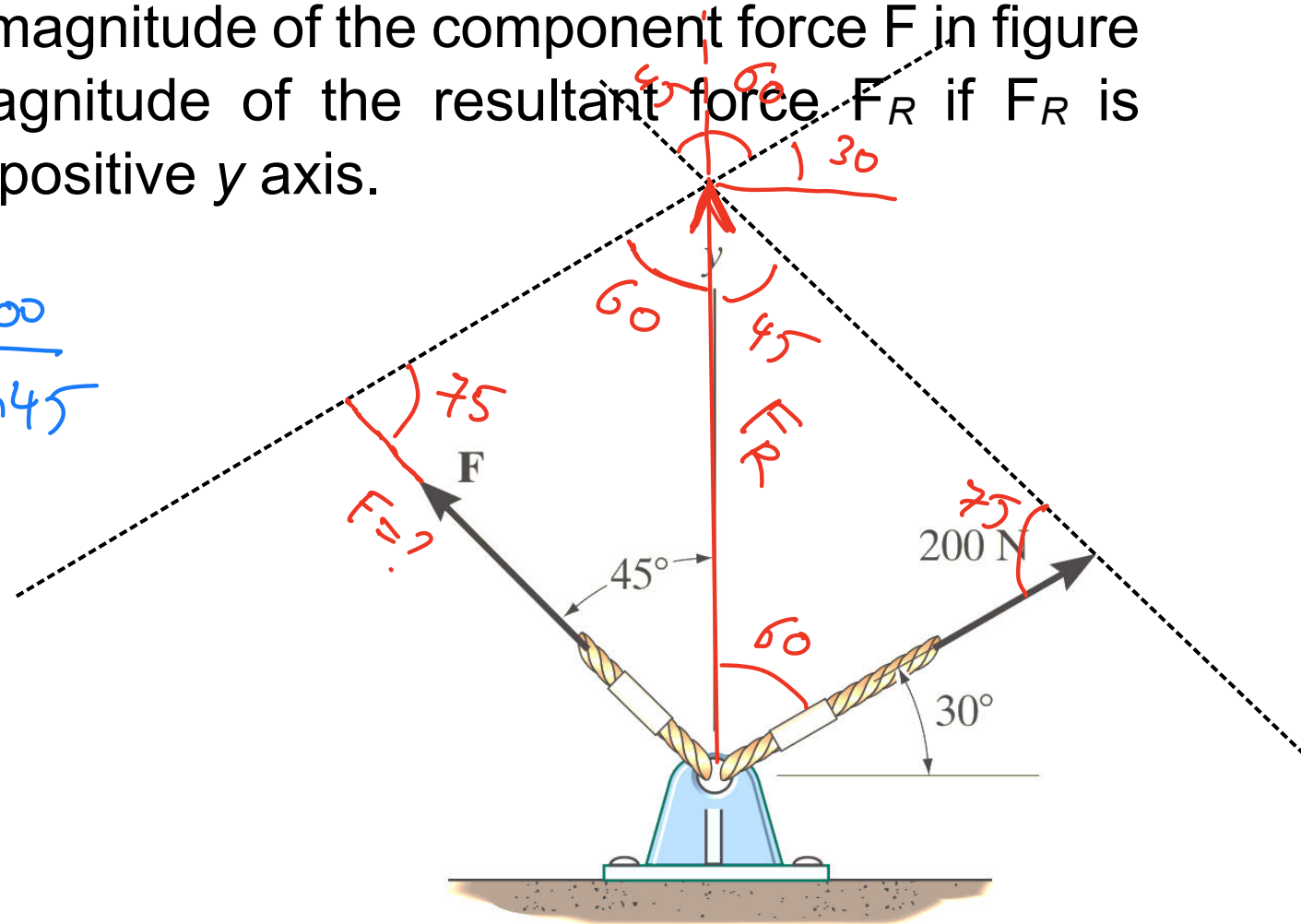


Practice at Home

2. Determine the magnitude of the component force F in figure below and the magnitude of the resultant force F_R if F_R is directed along the positive y axis.

$$\frac{F}{\sin 60} = \frac{F_R}{\sin 75} = \frac{200}{\sin 45}$$

$$200 \sin 30 = F \sin 45$$



Practice at Home

3. It is required that the resultant force acting on the eyebolt in the figure below be directed along the positive x axis and that $\mathbf{F}_2$ have a *minimum* magnitude. Determine this magnitude, the angle θ , and the corresponding resultant force.

